

RATON, NM, USA

WMO: 722678

Lat: 36.741N

Lon: 104.502W

Elev: 1935

StdP: 80.14

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 23052

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-15.5	-12.9	-22.5	0.6	-3.3	-20.2	0.8	0.8	15.7	2.0	13.9	3.6	3.0	50	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	33.8	14.8	32.4	14.7	30.9	14.6	17.7	27.1	17.1	26.5	16.6	25.9	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.7	19.5	14.4	13.0	19.1	13.7	12.5	18.9	57.4	27.2	55.4	26.1	53.5	25.6	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.5	9.8	-22.0	35.8	3.1	1.4	-24.2	36.8	-26.1	37.5	-27.8	38.3	-30.1	39.3		
(5)			-22.3	18.9	3.1	0.6	-24.5	19.3	-26.3	19.7	-28.1	20.1	-30.3	20.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	0.1	1.2	5.8	9.5	14.3	19.9	22.0	20.7	17.2	10.7	4.4	-0.4		
	DBStd	8.94	4.85	5.10	4.73	4.11	4.00	3.02	2.17	2.32	3.01	4.36	4.73	4.98		
	HDD10.0	1304	307	247	142	56	12	0	0	0	1	43	173	323		
	HDD18.3	3144	565	480	389	265	132	18	2	6	53	236	418	581		
	CDD10.0	1490	0	1	12	41	145	297	373	332	218	66	4	1		
	CDD18.3	289	0	0	0	0	7	65	117	80	20	1	0	0		
	CDH23.3	4560	0	0	11	43	329	1184	1438	1000	471	83	1	0		
	CDH26.7	1822	0	0	0	3	82	536	662	393	137	8	0	0		
(14) Wind	WSAvg	4.1	3.6	4.3	4.6	5.1	4.8	4.4	3.6	3.4	3.7	3.9	3.8	3.7		
(15) Precipitation	PrecAvg	408	12	9	20	32	43	52	75	73	41	26	13	9		
	PrecMax	605	47	37	53	140	144	102	138	163	99	104	42	33		
	PrecMin	275	0	0	0	7	0	7	16	16	4	3	0	1		
	PrecStd	89	13	10	16	30	36	34	35	36	25	23	11	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	20.3	24.7	26.7	31.2	35.9	35.2	34.4	31.9	27.8	22.7	17.1		
		MCWB	5.4	6.7	8.2	9.1	11.7	13.9	16.0	15.3	14.2	11.0	8.0	5.4		
	2%	DB	14.9	17.1	22.3	24.5	28.9	33.9	33.8	32.3	29.9	25.7	20.2	14.4		
		MCWB	4.2	5.1	7.3	8.5	11.1	13.4	15.7	15.4	13.8	10.5	7.1	4.2		
	5%	DB	12.6	14.8	19.7	22.6	27.1	32.0	32.5	30.7	28.3	23.6	17.6	12.0		
		MCWB	3.2	4.0	6.3	8.0	10.8	13.4	15.5	15.5	13.5	9.9	6.0	3.2		
	10%	DB	10.1	11.9	17.0	20.6	24.9	30.0	30.9	28.9	26.4	21.1	14.6	9.2		
		MCWB	2.1	3.0	5.4	7.3	10.2	13.4	15.6	15.3	13.1	9.5	4.8	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.1	9.0	11.3	15.0	17.6	18.6	18.5	16.8	14.7	9.1	6.2		
		MCDB	14.9	18.6	21.9	21.0	24.5	26.8	27.7	27.8	25.2	19.5	19.6	15.5		
	2%	WB	4.8	5.6	7.8	10.0	13.4	16.6	17.8	17.7	15.8	12.8	7.7	4.7		
		MCDB	13.8	16.2	20.4	20.0	22.6	26.3	27.4	26.9	24.0	19.9	18.4	13.1		
	5%	WB	3.6	4.3	6.8	8.9	12.2	15.7	17.2	17.0	15.1	11.3	6.5	3.4		
		MCDB	11.9	13.9	18.5	19.6	21.9	25.7	27.0	26.0	23.1	20.1	16.6	11.3		
	10%	WB	2.3	3.0	5.7	7.9	11.3	15.0	16.7	16.5	14.5	10.1	5.2	2.0		
		MCDB	9.7	11.2	15.9	18.8	21.6	24.9	26.5	25.1	22.0	19.1	14.2	8.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	16.8	18.0	17.6	17.5	18.3	17.2	16.4	16.8	17.6	17.6	16.0		
		MCDBR	20.9	21.8	22.7	22.0	21.0	21.4	19.9	19.0	20.2	21.9	21.7	19.7		
		MCWBR	12.0	11.6	10.8	9.8	8.4	7.6	5.9	6.1	7.2	9.5	11.0	11.3		
	5% WB	MCDBR	19.7	20.3	21.3	19.3	18.0	17.6	16.5	15.9	16.0	17.5	20.3	18.6		
		MCWBR	11.5	10.9	10.4	9.0	7.5	6.8	5.4	5.6	6.2	8.1	10.5	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.239	0.267	0.288	0.301	0.323	0.356	0.340	0.299	0.273	0.246	0.237		
	taud		2.630	2.632	2.547	2.493	2.518	2.455	2.443	2.474	2.556	2.601	2.607	2.606		
	Ebn at Noon		995	1023	1009	995	979	952	918	929	959	964	967	966		
	Edh at Noon		69	78	94	106	105	112	113	107	93	80	70	66		
(44) All-Sky Solar Radiation	RadAvg		3.08	4.01	5.26	6.40	7.14	7.46	6.90	6.35	5.64	4.45	3.34	2.68		
	RadStd		0.20	0.28	0.36	0.38	0.48	0.34	0.27	0.29	0.30	0.27	0.20	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	+46

Nomenclature: See separate page